

1. Construct a truth table for the compound proposition

$$(p \vee q) \wedge (p \vee \neg q) \leftrightarrow p$$

and express this proposition as an English sentence with the propositions "The integer n is even." and "The integer n is greater than 7." for p and q respectively.

	p	q	$\neg q$	$p \vee q$	$p \vee \neg q$	$(p \vee q) \wedge (p \vee \neg q)$	$(p \vee q) \wedge (p \vee \neg q) \leftrightarrow p$
Solution:	T	T	F	T	T	T	T
	T	F	T	T	T	T	T
	F	T	F	T	F	F	T
	F	F	T	F	T	F	T

As a sentence this statement reads that "The integer n is even if and only if the integer n is both even or greater than 7 and is even or less than or equal to 7"

2. Determine the negation of the proposition

"For every positive real number $x > 0$, there exists a positive integer n such that $\frac{1}{n} < x$."

and write out both the proposition and its negation using the quantifiers $\exists$ and $\forall$.

Solution: The negation is that "There exists a positive real $x > 0$ such that for every positive integer n we have that $\frac{1}{n} \geq x$. We can write these as $\forall(x \in \mathbb{R}, x > 0) \exists(n \in \mathbb{N})$ such that $(\frac{1}{n} < x)$ and $\exists(x \in \mathbb{R}, x > 0)$ such that $\forall(n \in \mathbb{N})(\frac{1}{n} \geq x)$.

3. Show that the proposition

$$p \leftrightarrow q$$

is logically equivalent to the proposition

$$(p \rightarrow q) \wedge (\neg p \rightarrow \neg q).$$

Make up propositions p and q and write this equivalence as an english sentence.

Solution: We have that $(p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$ as a contrapositive which is an expanded form of $p \leftrightarrow q$.

4. Consider the proposition

$$P(x) = \exists y (y^2 \leq x \leq y) \quad \text{for } y \text{ a real number.}$$

Can we determine for which real numbers x the proposition $P(x)$ is true?

Try to find another proposition $Q(x)$ so that $P(x) \equiv Q(x)$ where $Q(x)$ does not contain any quantifiers.

Solution: We have that $P(x) \equiv Q(x)$ where

$$Q(x) = "0 \leq x \leq 1"$$

This can be demonstrated by a proof by cases, i.e. taking the logical equivalence $P(x) \equiv (P(x) \wedge C_1(x)) \vee (P(x) \wedge C_2(x)) \vee (P(x) \wedge C_3(x))$ where $C_1(x) \vee C_2(x) \vee C_3(x)$ is tautological and each C_i represents a case of x being $x < 0$, $0 \leq x \leq 1$, or $x > 1$.